Pole Position from Thermally Cleaned Sibley Group Sediments and its Relevance to Proterozoic Magnetic Stratigraphy[1]

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Thermal cleaning of paleomagnetic samples of the Sibley Group shows that 3 directions of magnetization are present: a normal magnetization (240,+16), a reverse magnetization (048, −05), and a third direction (109,−69), which is parallel to that in the Logan Sills and presumed to have been acquired during their intrusion. The latter result confirms that the remanent magnetization of the Logan Sills was acquired at the time they cooled. Combining all relevant sites, a revised mean Logan Sills direction of 110,−73 yields a pole at 49N,138W, (dp = 7, dm = 7). The combined normal and reversed groups (232,+10) yield a pole position at 20S,214E. This result suggests that there is a bend in the North American polar wander curve between 1200 and 1400 m.y. ago. The curve is compared with the polar wandering curve derived from rocks of comparable age from southern Africa.

Le nettoyage thermique, fait sur des échantillons du Groupe de Sibley, montre 3 directions de magnétisation: une magnétisation normale (240, +16), une magnétisation inverse (048, −05), et une troisième (109, −69), cette dernière étant parallèle à celles trouvées dans les "Logan Sills" et donc, présumément contemporaine. Les touts derniers résultats montrent que le magnétisme rémanant des "Logan Sills" est contemporain de leur refroidissement. Utilisant les mesures pertinentes, on a trouvé (110, −73) comme direction moyennes des "Logan Sills". Cette détermination implique un pôle à 49N, 138W (dp = 7, dm = 7). En combinant les groupes normales et inverses; (232, +10) on obtient un pôle à 20S, 214E. Ces résultats nous portent à croire qu'il y a eu une courbure dans la migration du pôle de l'Amérique du Nord, il y a de ça 1200 et 1400 m.a. La migration du pôle ainsi obtenu est comparée avec celle obtenue sur des roches à peu près isochrone de l'Afrique du Sud.

Introduction

The polar wandering curve for North America is now broadly established back through time to 1200 m.y. ago (DuBois 1962, Robertson and Fahrig 1971, Spall 1971) and the trend of the curve is becoming evident even beyond 2000 m.y. ago (Irving and Park 1972). However, polar movement appears to have been quite slow between 1500 and 1200 m.y. ago and both the shape of the curve during this interval and also the variations in reversal frequencies of the earth's magnetic field need to be clarified. The Sibley Group is a useful unit for study of the magnetic field in this age range.

An earlier study of the Sibley Group by DuBois (1962) revealed three scattered groups of directions, and two samples baked by an overlying sill. The mean direction from his group II (reversed) (PDB II) is quite close to that for the Logan Sills (Fig. 1), and by no means opposite to his group I (normal) mean (PDB I): one may infer that some of his specimens were affected by baking. In this present work an attempt is made to understand the complex direction distribution by demagnetization studies, in order to determine an accurate pole for the Sibley Group.

Unweathered cylindrical specimens from a 1 in. (2.50 cm) oriented drill core were used in the sampling programme. The collection of suitable samples from Sibley Group sediments presents special difficulties. The rocks are predominantly very fine-grained, marly, and brittle, and are often so crumbly that solid samples are unobtainable. Where the rock has been baked, however, it is much better consolidated, and cores may be readily drilled. Hence a tendency exists for the sampling to be biased in favor of

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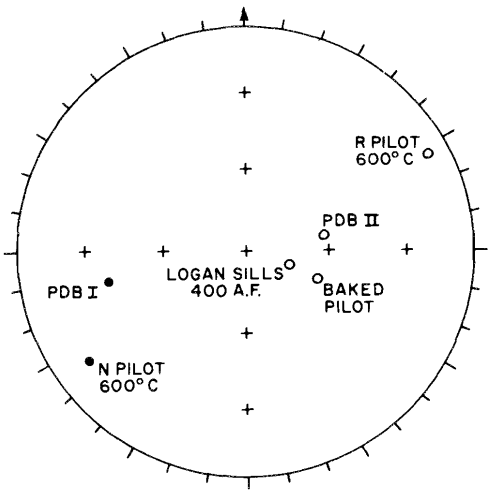


FIG. 1. An equal area stereographic projection showing the mean magnetization direction for the three main direction groups in the Sibley Group. The Normal and Reversed directions after heating pilot samples to 600 °C are given, and the initial baked direction is shown. They are compared with the mean direction from the Logan Sills (Robertson and Fahrig 1971), and DuBois (1962) directions (PDB I and PDB II) from the Sibley.

baked rock, although rocks were taken as far away as possible from known sills.

Geology and Age

The Sibley Group rests unconformably or disconformably on the Animikie Group, which consists in this area of shales of the Rove Formation above, lying conformably on taconitic, cherty, iron-rich rocks of the Gunflint Formation and the Archean crystalline basement. The Sibley Group is overlain disconformably by sediments and predominant basic lava flows of the Osler Formation.

Tanton (1931) divided the Sibley Group into 6 members. Recently Franklin (1970) modified Tanton's divisions into 7 units given below, (g) being the youngest.

(g) limestone
(f) purple mudstone*
(e) limy red mudstone*
(d) chert stromatolite
(c) sandy red mudstone*
(b) sandstone
(a) basal conglomerate

The units sampled are marked with an asterisk.

The mudstones of the Sibley Group are composed of authigenic clay (expandable montmorillonite–chlorite–vermiculite type), carbonate (dolomite–calcite), microcline, and hematite, and minor detrital quartz and feldspar (Franklin, personal communication 1971). The low degree of stability of the clay mineral, which breaks down at about 150 °C, may account for the extensive effect of the Logan Sills metamorphism which has affected the remanent magnetization (see bulk cleaning). The rocks are thought to have been deposited under fluvial or marine conditions, in a hot, dry climate (Franklin 1970), perhaps in a piedmont-valley flat type of environment on the edge of a rifting craton. The chert–stromatolite unit (d) is composed of "genuine index fossils . . . abundant in the Lower and Middle Riphean (Helikian)," (Hoffman 1969), which were formed in near tidal or intertidal zones. Surface outcrop of the Sibley Group is shown in Fig. 2. It is essentially flat-lying, and has not been regionally metamorphosed, but contains lead–zinc–barite vein deposits (Franklin 1970).

The rock units underlined in Table 1 are those for which the ages relative to the Sibley Group are known geologically. The Logan Sills intrude Sibley sediments and in places bake them, and thus are demonstrably younger. The Osler Volcanics visibly overlie the Sibley rocks, which in some places rest on Rove sediments, elsewhere on Gunflint rocks, and in places directly on the Archean.

Radiogenic ages are listed in Table 1. In general the radiogenic age sequence is an agreement with the stratigraphic sequence, although in some cases the uncertainties are such that overlap is possible and the relative radiogenic ages may not always be correct.

Alternating Field Demagnetization

Stepwise alternating field demagnetization of pilot specimens from each site revealed that a large initial scatter was not greatly reduced by any cleaning field (Table 2(b)). The mean directions for each cleaning field for the four groups that were differentiated by thermal cleaning are given in Fig. 3(a). The directions of all except the reversed group were but little affected by the alternating field treatment. It is

TABLE 1. Poles, ages, and succession

Formation	Pole position	A_{95}	Pole no. (Fig. 7)	Polarity	Geological setting	Radiogenic age	Radiogenic method	Remarks
Franklin igneous events	08 °N,167 °E	5	1	M	Dikes and sheets	675	K/Ar	Fahrig *et al.* 1971 interpretation of 30 K/Ar radiogenic ages
Freda Sst. and Nonesuch shale	09 °N,170 °E	4	2	N	Overlie Portage Lake volcanics			DuBois 1962
Portage Lake volcanics	24 °N,177 °W	9	3A	N	Overlain by Freda and Nonesuch			Vincenz 1968
Gila County sills	27 °N,179 °E	7	3B	N		1145	U/Pb K/Ar	Helsey and Spall 1972 Silver 1960, Damon *et al.* 1962
Upper North Shore volcanics	32 °N,173 °W	—	4	N	Overlie Reverse volcanics			Palmer 1970
Thunder Bay dikes	35 °N,179 °W	5	5A	N	Bake Logan sills intrude Rove Fm	1020	K/Ar	Robertson and Fahrig 1971 interpretation of 5 K/Ar radiogenic ages
Minnesota intrusives	35 °N,168 °W	2	5B and 6A	N	Duluth Gabbro intrudes North Shore volcanics	1115	Rb/Sr U/Pb	Fauré *et al.* 1969 Silver and Green 1963 Beck 1970
Mamainse Upper	37 °N,180 °E	4	6B	N				Palmer 1970, Van Schmus 1971
Lower North Shore volcanics	47 °N,160 °W	—	7	R	Underlie normal volcanics			Books 1968 Palmer 1970
Osler volcanics	49 °N,157 °W	9	8	R*	Overlie Sibley Group			Tanton 1931, Palmer 1970
Mamainse Lower	54 °N,148 °W	8	9A	R				Palmer 1970, Van Schmus 1971
Logan sills	47 °N,140 °W	8	9B	R	Intrude Sibley and Rove	1050	K/Ar Rb/Sr	Robertson and Fahrig 1971 interpretation of 8 K/Ar radiogenic ages confirmed by Rb/Sr isochron
Abitibi dikes	27 °N,134 °W	5	10	N	Intrude Huronian	1150	K/Ar	Larochelle 1966 See Robertson and Fahrig 1971
Mackenzie igneous events	04 °N,171 °W	5	11	N	Volcanics, dikes, and layered intrusion	1200	K/Ar Rb/Sr	Fahrig and Jones 1969 Robertson 1969 interpretation of 35 K/Ar radiogenic ages with confirmatory Rb/Sr isochron

TABLE 1 (*Concluded*)

Formation	Pole position	A_{95}	Pole no. (Fig. 7)	Polarity	Geological setting	Radiogenic age	Radiogenic method	Remarks
Sibley Group	20 °S,146 °W	9	12	M	Overlies Rove Formation	1370	Rb/Sr	R. K. Wanless personal communication. Tanton 1931. Franklin 1970
Front Range intrusives	08 °S,151 °W	4	13A	M	Red granite, basic inclusions, and andesite dikes	1410	Rb/Sr	Peterman *et al.* 1968 Eggler and Larsen 1968 Ferris and Krueger 1964
Belt Series of Montana combined	09 °S,151 °W	12	13B	M	McNamara, Miller Peak, Spokane, Grinnel, Appekunny			Irving 1964
Michikamau anorthosite	01 °S,145 °W	6	14	N		1400	K/Ar	Murthy *et al.* 1968 Emslie 1964
Croker Island complex	05 °N,143 °W	9	15	N		1475	Rb/Sr	Van Schmus 1965 Palmer 1969
Rove Formation	—	—	—	—	Overlies Gunflint Formation	1664	Rb/Sr	Peterman 1966
Gunflint Formation				M	Rests on basement			

NOTES: The pole positions, numbered in the table, are shown on Fig. 7. The formations italicized are those known from stratigraphic or contact relations to have relative ages as shown in the table.

The polarity shown is either Normal (N), that is to say if the polar path is traced back from the present pole, the direction of magnetization is in the same sense as the present field, or Reversed (R), in which case it is in the opposite sense from the present field, or Mixed (M), for formations in which both polarities have been observed.

No errors are given with the radiogenic ages, which are quoted in millions of years, as the laboratory standard deviation usually quoted takes no account of other possible sources of error, such as argon loss and contamination, and may be misleadingly low. However the reliability of the quoted ages covers a wide range; for estimates of this reliability the original papers describing the radiogenic work should be consulted.

Where the radiogenic age is based on Rb/Sr isochron, the number is based on a decay constant $\lambda = 1.39 \times 10^{-11}\ yr^{-1}$. Where the workers used a decay constant of $\lambda = 1.47 \times 10^{-11}\ yr^{-1}$ the quoted age has been correspondingly adjusted.

The Duluth Gabbro radiogenic age, which is firmly based, does not fit the age suggested for this part of the curve. This is a problem; one possibility is that the K/Ar radiogenic ages are too low, as a result of argon loss, in which case the age of the curve would be older than it is shown, (*e.g.* Gates 1971).

*Palmer (1970) did not sample the highest levels of the Osler Volcanics, and the aeromagnetic maps indicate that they may be normally magnetized. If this is so it does not necessarily mean that the Osler Volcanics formed during a Mixed Polarity Interval, but more likely that they were extruded through the time of the transition of polarity between the lower (older) Reversed, and upper (younger) Normal Intervals.

The validity of the comparison of poles from formations of normal, mixed, and reversed polarity is based on the assumptions that secondary components have been removed (Palmer 1970) and that the field in which they became magnetized was that of a geocentric axial dipole (Wilson 1970).

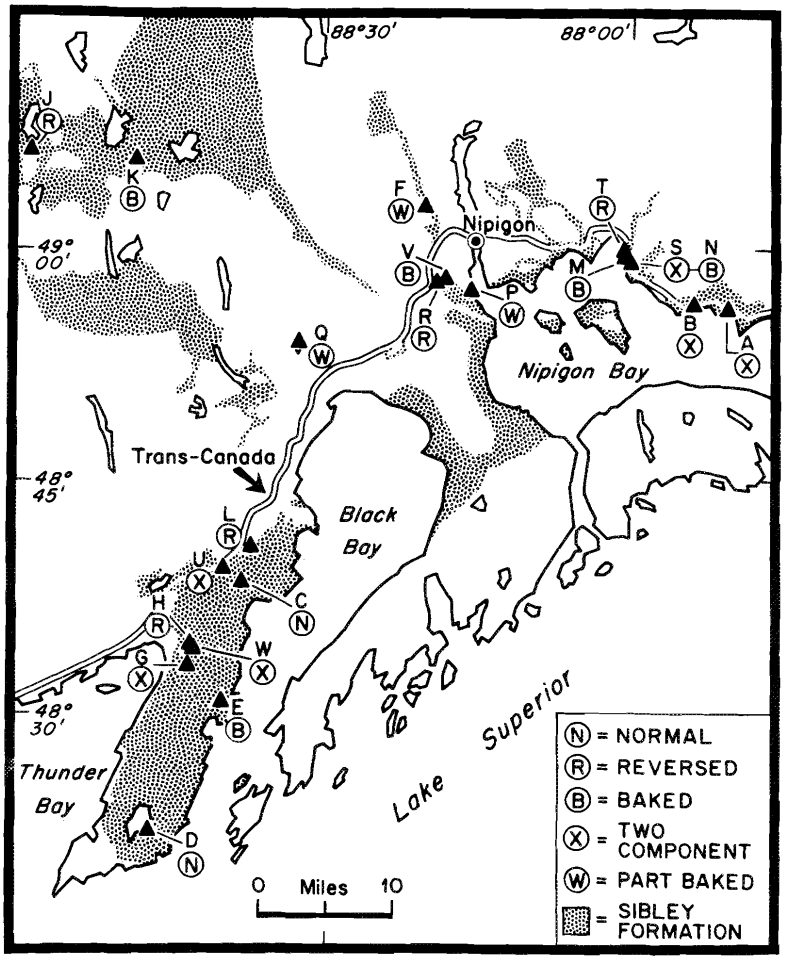


FIG. 2. A geological sketch map showing the extent of the Sibley Group and indicating the sampling sites and whether the directions of magnetization obtained from them are normal (N), reversed (R), baked (B), two component (X), or partly baked (W). The site letters refer to the sites shown in Table 3.

noteworthy that the partly baked specimen displays a stable end point in the 300–1000 Oe range in a direction close to the mean for DuBois' Group II.

The demagnetization curves of specimens from the normal group (Fig. 4(a)) and the baked group (Fig. 4(c)) indicate that the specimens have a very high coercivity. Erratic variations in intensity from reversed sites R and H (Fig. 4(b)) are thought to be due to the interaction of two vectors (along the present field and the reversed Sibley field) of widely different direction that tend to oppose each other and yield a small residual intensity. It is concluded that this technique is unsuitable for obtaining the original direction of magnetization of the Sibley Group.

Thermal Demagnetization of Pilot Specimens

The large scatter of directions of NRM can be explained as the interaction of four vectors, and a measuring error, inflated when working near the limit of sensitivity of the magnetometer.[2] The four vector components, different

[2]The astatic magnetometer (Roy *et al.* 1972*a*), is sufficiently sensitive to measure intensies down to 1×10^{-7} e.m.u.; however, in some instances, the magnetization induced in largely demagnetized specimens by the magnetic field of the magnet may obscure the true direction.

TABLE 2. Pilot demagnetization

(a) Thermal

Treatment temperature	Normal (4 specimens)			Reversed (3 specimens)			Baked (5 specimens)			Partly baked (1)	
	D	*I*	*k*	*D*	*I*	*k*	*D*	*I*	*k*	*D*	*I*
NRM	251	+13	11	017	+06	2.6	111	−74	14	027	−63
480 °C	265	+12	8	061	−15	6	143	−67	23	003	−59
520 °C	258	+11	16	059	−25	2.8	142	−74	27	065	−57
560 °C	256	+14	12	054	−15	9	155	−70	16	062	−53
600 °C	235	+17	36	062	−09	13	138	−65	43	056	−36
640 °C	255	+18	2.3	064	+02	132	143	−66	29	057	−31
660 °C	231	+16	40	063	−01	525	151	−67	10	052	−32
670 °C	214	−05	28	069	−07	36	156	−64	12	054	−27
680 °C	231	+16	18	066	−06	1300	163	−59	6	053	−25
700 °C	207	+03	—	069	+13	—	240	−24	1.8	061	−01

(b) Alternating Field

Treatment peak field	Normal (6 specimen)			Reversed (4 specimens)			Baked (8 specimens)			Partly baked (1)	
	D	*I*	*k*	*D*	*I*	*k*	*D*	*I*	*k*	*D*	*I*
NRM	277	+42	6	051	+08	2.3	084	−66	4.3	032	−59
100 Oe	275	+41	10	353	−15	1.4	115	−68	4.2	051	−64
200 Oe	275	+40	8	018	+01	2.3	115	−75	15	054	−63
300 Oe	276	+42	8	341	−24	1.1	119	−78	13	065	−62
400 Oe	268	+35	7	344	−10	1.2	118	−76	16	064	−61
600 Oe	280	+33	9	043	−22	1.7	138	−76	10	066	−55
1000 Oe	281	+37	9	311	−45	1.3	129	−77	15	063	−59

NOTES: The dashed lines indicate results in which one or more specimens were too weakly magnetized to yield a result. *D* is the declination, clockwise from true north. *I* is the inclination, positive downwards. *k* is Fisher's (1953) best estimate of precision.

proportions of which make up the NRM of the specimens from the Sibley Group are as follows:

(a) A viscous component along the present earth's field direction at 000,+77.

(b) A component formed by baking by over- or underlying Logan Sills magnetized in a direction of about 107,−72;

(c) A reversed component, acquired when the sediments were deposited, in a direction of about 048,−05 (Table 4).

(d) A normal component, also acquired when the sediments were deposited, in a direction of about 240,+16, (Table 4).

Thermal demagnetization, using the oven of Roy *et al.* (1972*b*), of a pilot specimen from each site revealed that after high temperature thermal cleaning the sites separated into the 5 groups of Table 3 (see Table 2(a) and Fig. 3(b)). Directions from the three main groups are shown in Figs. 3(c) and (d) before and after thermal cleaning respectively. The fourth and fifth groups contain sites in which two or more of the four components appear to be so intimately mixed that no known treatment removes one component without also removing a significant portion of the other.

Thermal demagnetization in the temperature range 480°–670 °C revealed that the effect of heating to temperatures within this range differed from site to site with respect to the proportion of components remaining. Typical demagnetization curves in this range are shown in Fig. 6. It became necessary as a result of this to make bulk runs for different sites at different temperatures. Some interesting features of direction changes after these bulk runs are shown in Fig. 5. The magnetization became so weak for specimens from some sites (for instance sites B and T) that it was impossible to obtain measurements after heating to 600 °C, but it is evident from the results for these two sites that after heating to the highest temperature for which results could be obtained (480 °C for site T and 560 °C for site B), a large (secondary) component in a direction

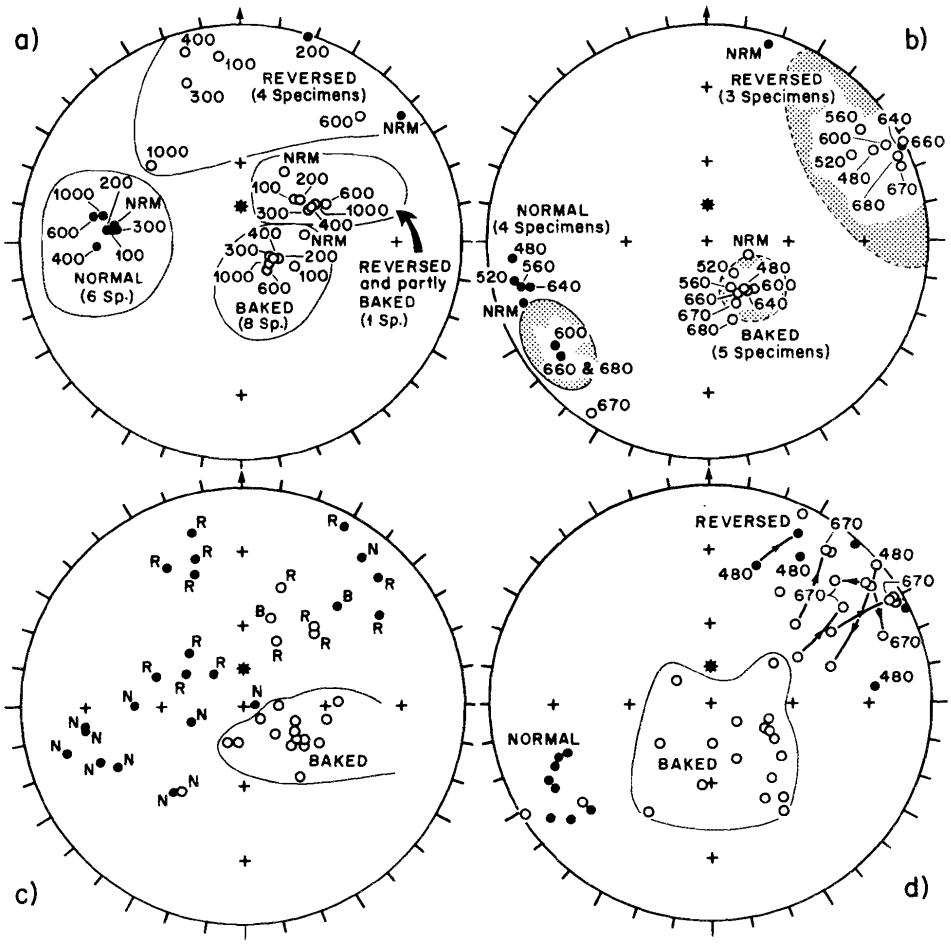


FIG. 3. Equal area stereographic projections showing:
(a) Mean directions from pilot specimens for the different groups in the Sibley Group after successive alternating field cleanings up to a peak field of 1000 Oe. Note how close the 'partly baked' direction falls to the mean of DuBois (Fig. 1 PDB).
(b) Mean directions from pilot specimens from the three main groups after heating to the temperature indicated in degrees centigrade. Ovals of confidence at $P = 0.95$ are also shown for each group after the 600 °C heating. These are larger (less accurate) than the final normal and reversed confidence ovals as they are based on fewer samples.
(c) Initial specimen direction for all specimens included in the normal (N), reversed (R), and baked groups.
(d) Directions for the specimens of Fig. 3(c) after thermal cleaning at a temperature of 560 °C, except where a different temperature is indicated. The dipole field direction is shown as an 8 point star.

near the present earth's field was much reduced relative to a reversed and normal component respectively.

At first sight (Fig. 5) it would seem that sites C, D, H, and T initially contain a viscous component in the present dipole field direction that is preferentially removed by heating. This simple explanation does not fit the initial directions at sites B, L, and R, however, nor the fact that the spurious component is not greatly affected by heating to 200 °C, a temperature high enough to remove viscous components (Thellier and Thellier 1959). A more likely explanation for the scattered directions is that these rocks contain both a normal and a reversed component, possibly one detrital and the

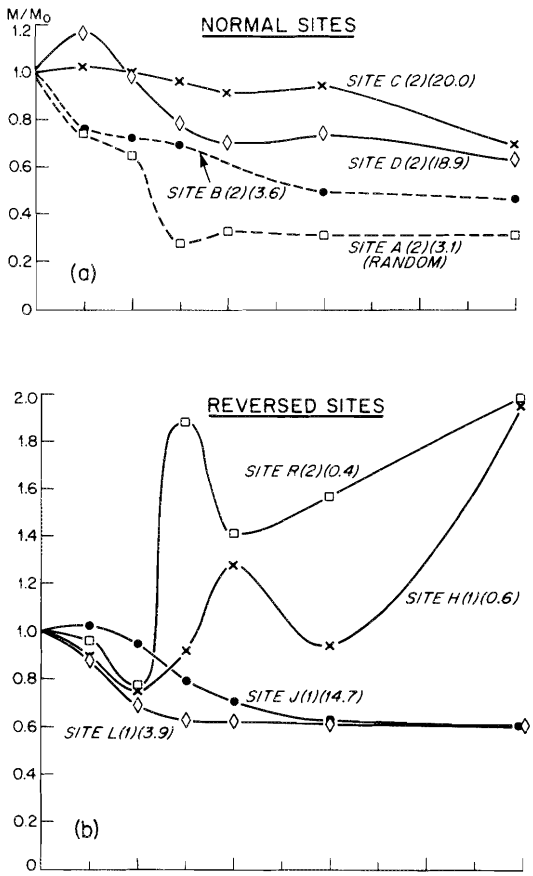


FIG. 4. Typical alternating field demagnetization curves for: (a) the normal (N) group and also a random site, (b) the reversed (R)—note that the erratic intensity variation occurs at very weakly magnetized sites, (c) the group in which the rocks were baked by nearby Logan Sills (B). The first bracketed figure is the number of specimens used to define the points. The second bracketed figure is the initial remanent magnetization intensity in emu $\times 10^{-6}$.

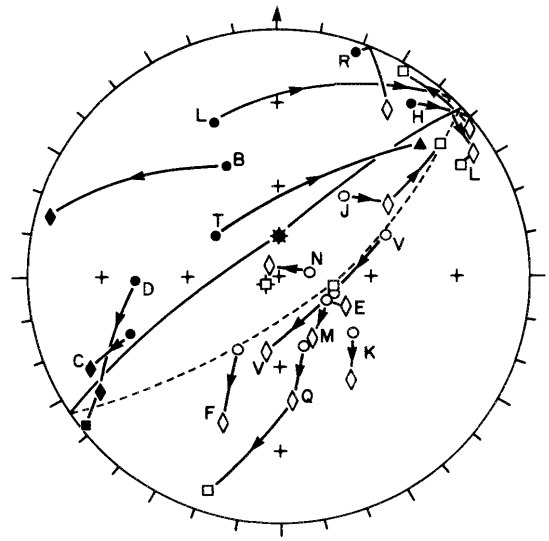


FIG. 5. An equal area stereographic projection showing the mean direction changes after heating 3 or more specimens from the sites indicated. Some partly baked sites have been omitted for clarity. The temperature to which the groups were heated is indicated as follows: ● = 20 °C, ▲ = 480 °C, ◆ = 560 °C, ■ = 670 °C. The mean direction for all the Sibley Samples (-□-) (Robertson and Fahrig 1971) and the dipole field directions (✸) are also given. The great circle through the normal, reversed, and baked directions, and the dipole field direction, is shown.

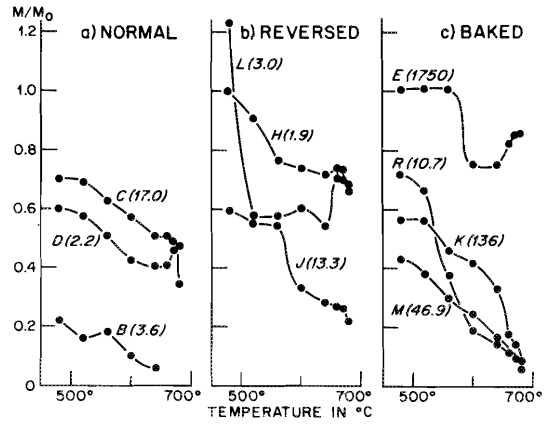


FIG. 6. Thermal demagnetization curves in the directionally interesting range of 480–680 °C for typical specimens from (a) the normal group, (b) the reversed group, and (c) the baked group. The letters refer to the sites of Fig. 2 and the bracketed figures to the natural remanent intensity (M_0) in e.m.u. $\times 10^{-6}$.

TABLE 3. Mean site statistics

Site position			Initial directions				Directions after heat treatment				
Site	Lat.	Long.	N	D I	k	α	Temp.(°C)	N	D I	k	α
Normal											
C	48.6N,	88.6W	5	248, +36	27	15	560	5	243, +17	31	14
D	48.4N,	88.8W	5	268, +52	1.3	130	560	5	236, +15	11	24
Reversed											
H	48.6N,	88.7W	2	37, +12	33	45	560	2	51, −2	32	46
J	49.1N,	89.0W	3	39, −55	89	13	670	3	50, −16	28	24
L	48.7N,	88.6W	3	338, +32	188	9	670	3	58, −15	31	23
R	48.9N,	88.3W	3	19, +03	2.6	103	560	3	32, −19	17	31
T	49.0N,	88.0W	4	313, +65	72	11	480	4	46, +22	7	37
Baked											
E	48.5N,	88.7W	4	118, −72	124	8	560	4	115, −66	583	4
K	49.5N,	88.8W	4	128, −59	131	8	560	4	146, −47	154	7
M	49.0N,	88.0W	3	109, −71	19	29	560	3	151, −67	19	29
N	49.0N,	88.0W	3	84, −80	7	51	560	3	316, −86	8	45
V	48.0N,	88.2W	3	68, −52	1.5	—	560	3	189, −64	14	35
Two component											
A	48.8N,	87.8W	5	270, +63	8	28	560	5	338, +23	2.4	63
B	48.8N,	87.9W	4	335, +49	13	27	560	4	285, +6	7	36
G	48.5N,	88.7W	6	294, +29	7	27	560	6	331, +26	1.8	74
S	49.0N,	88.0W	4	170, +74	6	42	560	4	238, 0	1.7	110
U	48.7N,	88.7W	4	321, +79	8	35	480	4	321, +68	4.7	48
W	48.6N,	88.7W	5	282, +74	6	34	560	5	150, +59	4.0	44
Part baked											
F	49.1N,	88.4W	2	209, −62	30	47	560	2	201, −35	68	31
P	48.9N,	88.3W	3	227, +16	1.1	—	560	3	107, −80	1.2	—
Q	48.8N,	88.6W	4	160, −65	106	9	670	4	198, −7	22	20

NOTES: N is the number of separately oriented samples. D, I, and k have the same significance as in Table 2. α is the half angle of the cone of confidence at $P = 0.95$ (Fisher 1953). The site letters correspond with those of Fig. 2.

other diagenetic as suggested by Roy and Robertson (1968) for the Maringouin Formation. This explanation implies at least 2 reversals during the deposition of the Sibley Group, since the component remaining, in each case, must have the same blocking temperature and is likely to have the same origin. One transition would then come between the times of deposition and diagenesis of sites B, C, and D (normal sites when cleaned) and the other between the deposition and diagenesis times of sites H, L, and R (reversed sites when cleaned).

Stepwise heating of specimens for sites J and Q showed that initial baked components were progressively reduced as the heating temperature was increased in steps up to 670 °C, and the directions moved towards those for the Sibley reversed and normal groups respectively. In the case of site J the direction obtained after the 670 °C heating indicates that the baked component has been erased, so that this direction reveals a pre-Logan Sills intrusion field direction (reversed), probably that at the time the rocks formed. It was not possible to eliminate completely the effect of baking at site Q (Fig. 5) which is therefore placed in the part baked group (Table 3). Sites F, M, and K appear to be similar to site Q in that they were originally normally magnetized, but the baking was more extensive, and site E shows the extreme case of baking, in which all of the original component has been obliterated.

How Many Reversals?

Although both normally and reversely magnetized sites were found in the Sibley Group, none were found of mixed polarity. If only one field reversal is present the boundary between normally and reversely magnetized rock could provide a valuable marker horizon. Unfortunately in the flat-lying strata the relative stratigraphic position of sites is not clear and it was not possible to determine how many reversals are represented, and if only one which polarity is older. Detailed sampling of a 30 m section at site T, in an attempt to find a polarity change in a section, revealed a thickness of 9 m of reversely magnetized rock over- and underlain by weakly magnetized rock that gave scattered directions. This study only shows that

there is at least one polarity reversal during the time that the Sibley Group acquired its magnetization.

The Extent of the Baking

Some sites (E, K, M, N, V, F, P, Q) were baked, to varying degrees, by intruding Logan Sills, but the sills were not exposed at the sites, so that it was not possible to tell how close they were to the samples, nor whether at the time of cooling they had been above or below the sediment. To study this problem a roadcut exposing some 30 m of sediments, capped by a sill 1.3 m thick, and intruded 3 m lower in the section by a sill 0.5 m wide, was sampled. Pairs of samples were taken, where possible, at 2 m intervals below the sill. Within 2 m of the lower sill the sediments were too shattered to sample. At the 2 m level and further from the sill the dolomite and mudstone was very weakly magnetized and gave no magnetic indication of baking. Doell (1972) found that baking of country rock by a dike does not extend out from the margin by more than its thickness. This rule seems to apply below the sill, but heating above may be more extensive.

Bulk Cleaning

Pilot cleaning showed that the samples could be separated thermally into 5 groups and thermal cleaning of all samples was carried out. Sites T and U were cleaned at 480 °C because at higher temperatures they became too weak to measure. All other sites were cleaned at 560 °C. Sites J, L, and Q, which gave partly strung distributions at 560 °C, were further cleaned at 670 °C. Results are given in Table 3.

Sites A, B, G, S, U, and W appeared to contain a secondary component which could not be completely eliminated by the heat treatment before within-site directions became random (Table 3). These sites were put into a separate group, described as two component, and excluded from the pole position calculations. Sites F, P, and Q (part-baked) contain a component parallel to that in the Logan Sills, which also could not be completely eliminated by the thermal cleaning. They are also excluded from the pole position calculations. The boundaries between part baked and two component sites and unaffected sites is gradational, and it can be seen in Table 3 that sites V and F are marginal to the part baked and baked groups respectively.

The end-point criterion (As and Zijderveld 1958, Gough and Brock 1964) has been used to determine the best temperature for cleaning (see Robertson and Baragar 1972), and although the within-site scatter may be quite large after high temperature cleaning, the change in mean site direction is small in the end-point region (see Table 2).

The mean directions and statistics for the three main groups before and after thermal cleaning are given in Table 4. The means for both the normal and reversed groups, taken either by sites or by samples (Table 4) are markedly more precise after thermal cleaning, and the amount by which the normal and reversed directions depart from a perfect reversal is reduced by cleaning from 89° to 16°.

Discussion

This pole position for the Sibley Group is 36° away from that obtained by DuBois (1962), and the most likely source of this difference is the retention of secondary components acquired in more recent fields, rotating the DuBois directions. In this case one later field direction has been identified, that acquired by the Logan Sills at the time of their intrusion, and another inferred, that of opposite polarity soon after deposition.

Some sites, notably B, L, and T, gave initial directions (see Fig. 5) close to the present field direction, thought to be caused by viscous components. Such viscous components are readily removed by thermal cleaning (Thellier and Thellier 1959).

To obtain a reliable radiogenic age relevant to a paleomagnetic direction it is desirable to get concordance between two or more radiogenic methods which use different elements that are unlikely to have been affected by later events in the same way. This is often difficult as each method requires the presence of a different mineral assemblage. Only the Duluth Gabbro, the Logan Sills, and the Mackenzie Igneous rocks of the succession in Table 1 fulfil this reliability criterion.

Wherever in geological time polar wandering is continuous (*i.e.* quasistatic intervals (Briden 1967) must be excluded), it is possible that

TABLE 4

Group and sites	Untreated					Thermally cleaned							
	Directions					Directions					Pole positions		
	N	*D* *I*	*k*	α	*M*	*N*	*D* *I*	*k*	α	*M*	Lat. Long.	*dp*	*dm*
Baked Group (Sites E, K, M, N, V)													
By sites	5	101, −69	28	15	350	5	151, −69	17	19	325	71N, 206E	28	33
By samples	17	109, −69	8	14	409	17	147, −68	13	11	381	69N, 199E	15	18
Normal Group (Sites C, D)													
By sites	2	257, +44	29	48	10.5	2	240, +16	267	15	5.2	13S, 210E	8	16
By samples	10	253, +41	2.8	36	10.4	10	240, +16	18	12	5.1	13S, 210E	6	12
Reversed Group (Sites H, J, L, R, T)													
By sites	5	11, +15	2.4	63	4.3	5	48, −06	17	19	1.6	24S, 218E	10	19
By samples	15	4, +21	2.4	33	4.1	15	48, −05	10	13	1.5	24S, 218E	7	13
Grand Averages (N + R)*													
By sites	7	209, +07	2.1	57	6.0	7	231, +09	22	13	2.7	21S, 216E	7	13
By samples	25	209, +06	1.7	33	8.6	25	232, +10	11	9	2.9	20S, 214E	5	9

NOTE: Column headings have the same meaning as those of Table 3. M is the mean intensity in e.m.u. $\times 10^{-6}$. *dp* is the semi-axis of the oval of confidence along the great circle passing through the formation, at a probability $P = 0.95$, and *dm* is the semi-axis perpendicular to *dp*.

*The directions and pole positions of the reversed group have been reversed so that they form a group with the normal group in the grand averages.

relative ages between key points, such as the Mackenzie Igneous Events, may be best indicated by the relative positions of the poles on the polar wandering path, if no within-shield relative movement has taken place. Furthermore this paleomagnetic estimate of relative age may be sharpened by examination of the reversal pattern.

The pole position for the Sibley Group (Table 4) falls in the South Pacific, farther south than poles obtained from immediately older and younger rocks of the Canadian Shield (Fig. 7). Although the reversed group pole is 13 °SE of the normal group one, the latter is southeast of the belt pole, and the preponderance of reversed (15) to normal (10) samples in the grand average (Table 4) is not a significant factor causing the Sibley pole to fall southeast of the somewhat older poles (13–15) shown in Fig. 7.

The Sibley Group pole position derived here (215E, 20S) places the sediments themselves very close to the paleo-equator (within 10°) (See Fig. 7). This is consistent with the formation of expandable clays (corrensite), the red color, and the widespread development of dolomite and stromatolites (Hoffman 1969).

The tentative polar path for the Canadian Shield from 1500 m.y. to 700 m.y. ago shown in Fig. 7 forms a double loop, but the middle limb is poorly defined. Pole positions from formations of ages between that of the Logan Sills and the Sibley Group are clearly necessary. It is possible that the movement along the curve is caused by two elements, one a slow drift of the continent to the northwest, and the other, superposed on this, three periods of relatively rapid polar wandering almost at right angles (*cf.* Irving and Robertson 1969). Steady drift from Croker Island time (1475 m.y.) to Franklin time (675 m.y.) would give an average movement of 0.7 cm/yr, whereas polar movement between Mackenzie time (1200 m.y.) and Logan Sills time (1050 m.y.) would give a minimum rate (since the relative movement may all have taken place during only a part of that time) of 3.8 cm/yr.

The polar wandering curve for the African shield during the Proterozoic (Brock and Piper 1972) gives a criterion with which to compare the Canadian shield curve. The younger part of the African curve forms a great open bow in the same sense as the Logan Loop. The older side of this African bow is radiogenically dated at somewhat younger than the corresponding side of the Logan Loop, but uncertainties in dating allow that they may be contemporaneous. If this is so the difference in the separation of the sides of the Canadian loop and the

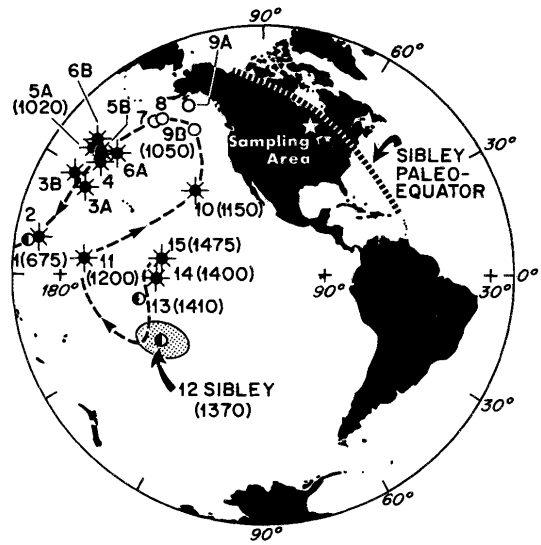


FIG. 7. An equal area stereographic projection showing the equatorial face of the earth from 160 °E to 20 °W. A suggested polar wandering curve for North America for the Upper Paleohelikian, Neohelikian, and earlier part of the Hadrynian Eras is given. The numbering of the poles is by age, one being the youngest, as far as is known; the age relationships used are a combination of stratigraphic control, radiogenic age dating, and the position of the pole on the curve. The poles shown are: 1. Franklin (Fahrig *et al.* 1971), 2. Freda and Nonesuch Shales (DuBois 1962), 3A. Portage Lake Lavas (Vincenz 1968), 3B. Gila County Sills (Helsley and Spall 1972), 4. Normal North Shore volcanics (Palmer 1970), 5A. Thunder Bay dikes (Robertson and Fahrig 1971), 5B and 6A. Minnesota intrusives (Beck 1970), 6B. Mamainse Upper (Palmer 1970), 7. Reversed North Shore volcanics (Palmer 1970), 8. Osler Lavas (Palmer 1970), 9A. Mamainse Lower (Palmer 1970), 9B. Logan Sills (Robertson and Fahrig 1971), 10. Abitibi dikes, (Larochelle 1966), 11. Mackenzie igneous events (Fahrig and Jones 1969), 12. Sibley Group (this paper), 13(A). Front Range (Eggler and Larson 1968), 13(B). Belt Series, Montana, combined (Irving 1964), 14. Michikamau Anorthosite (Murthy *et al.* 1968), and 15. Croker Island Complex (Palmer 1969). The oval of confidence at $P = 0.95$ for the Sibley Group is also shown. Stellar, half-moon, and open poles are derived from formations that are of normal, mixed, and reversed polarity respectively.

African bow may be interpreted as relative movement of the two shields. The lower (older, Fig. 7) loop of the Canadian Shield curve covers a time span (1400–1200 m.y.) in which data from both continents is sparse, but Brock and Piper's (1972) curve for Africa indicates again a more open loop, also suggestive of relative movement of the two shields in the same sense as in the younger part.

Since the earth's magnetic field is worldwide its polarity at any time must be the same all over the world. Hence any time span known to contain normally (reversely) magnetized rocks from the Canadian Shield must also contain normally (reversely) magnetized rocks in Africa and all other shields. Also periods (like the Tertiary) in which the polarity of the field alternated frequently from one polarity to the other should appear in the rock record as contemporaneous periods of mixed polarity on all the continents.

The polarity of Proterozoic formations considered to be well-dated either radiogenically or stratigraphically from the Canadian Shield is shown in Fig. 7 and Table 1. A pattern is evident. A normal polarity interval before 1400 is followed by one of mixed polarity. This is followed by normal polarity through the time of intrusion of the Mackenzie igneous events and the Abitibi dikes, then comes a reversed interval during which the Logan Sills and Lower North Shore and Osler volcanics were intruded. A normal interval followed, in which Thunder Bay dikes were intruded, Portage Lake lavas extruded, and the Freda and Nonesuch shales laid down. Towards the close of the Precambrian the Franklin Magmatic activity took place in a mixed polarity interval.

Large gaps in the record, for instance between Sibley and Mackenzie time, leave the pattern open to add additional polarity intervals, but all will apply equally to all the continental shields. Present paleomagnetic data from Africa does not appear to include any rocks formed during the reversed interval represented in Canada by poles 7, 8, and 9 (Fig. 7). A search in Africa for rocks formed in this reversed interval should help to define, and maybe deepen, the apex in the bow in the African polar-wandering curve, and make comparison of a contemporaneous portion of the curves from the two continents possible.

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